

source initiative transition to a closed-source model in the future.

Open source in the age of AI and generative AI

Open source AI technologies, including recent LLMs such as Google's Gemma, Meta's Llama, Alibaba's Qwen, DeepSeek's V3, and Microsoft's Phi, are rapidly advancing. These models are continuously narrowing the gap in effectiveness and performance compared to proprietary foundational models.

Since AI relies on large-scale data processing, the cloud remains the preferred choice for running these models due to the significant infrastructure requirements, including GPU-intensive workloads. However, conservative institutions like banks and government organisations often seek on-premises solutions. Open source models such as Mistral and Llama provide viable alternatives for these needs.

Efforts by platforms like Ollama further enhance accessibility by containerising LLMs, making them available for local execution across various operating systems, including Windows and macOS. Ollama simplifies deployment and installation while offering user-friendly APIs for seamless integration.

Some key advantages of running models locally through Ollama include lower latency and enhanced data privacy. A unique benefit is the consolidation of multiple open source LLMs under one framework. However, it's important to note that these models may not always be the latest versions and often represent optimised or reduced-scale variants of broader foundational models. For applications where the scope of AI requirements is well-defined, these models can be a practical and efficient solution.

These models can also be conceptualised as small language

models (SLMs), focusing on narrower use cases and leveraging optimisation techniques such as model quantization. This technique enables more efficient storage of parameters by reducing precision, utilising fewer bits instead of full floating-point representations.

Overall, open source AI is not merely catching up—it is actively leading the way, catering to organisations that prefer alternatives to cloud-based LLMs. By providing local execution capabilities, these solutions empower institutions to harness AI's potential without compromising on control and security.

Summarising the journey

Since its inception, open source technology has evolved in a highly organic yet strategic manner, steadily earning the confidence of the broader ecosystem. This growth has been fuelled by the collaborative efforts of academia, industry leaders, research institutions, and global developer communities. Today, open source functions as a parallel innovation engine, advancing in step with transformative technologies such as generative AI.

Over time, open source has matured into a credible and secure alternative to proprietary solutions—particularly for organisations operating under strict regulatory and compliance frameworks. It has consistently demonstrated its ability

to meet enterprise-grade security and governance standards, making it a compelling choice for those with concerns around processing sensitive or personally identifiable information (PII) in public cloud environments.

Importantly, the strong governance structures and mature operating models surrounding open source platforms significantly mitigate the risk of disruption—even in the event of a transition to a closed source model. These mechanisms ensure continuity, maintainability, and long-term sustainability, providing leadership with the assurance that their strategic investments will remain resilient and viable, regardless of shifts in licensing or ownership.

For organisations navigating the complexities of data privacy and cloud adoption, a hybrid architecture offers a practical and strategic path forward. By hosting non-sensitive workloads in the cloud while retaining sensitive data on-premises, enterprises can strike a balance—leveraging the scalability and innovation of cloud platforms while preserving full control over critical data assets. This hybrid approach supports both agility and governance, aligning with long-term business and technology goals.

Are organisations truly resilient enough to adapt if key open source technologies were to shift towards closed source models? That's something to reflect on. **END** 🐼

References

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